Term Final Examination, Autumn 2024
Course Title: Data structures and Algorithms I
course code: CSE 211

University of Information Technology & Sciences (UITS)
Faculty of Science and Engineering
Department of Computer Science and Engineering
Program: B.Sc. in CSE
Term Final Examination, Autumn 2024
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Marks: 50 Time: 3*(three)* hours

(Answer all questions)

1. **a)** Demonstrate the algorithm to pop an element from a stack. Consider the FIFO data structure, supporting two operations addition and deletion for the following sequence: **[05]**

M N O # P Q R # S T # U ! V W # # X !

For this sequence, interpret the sequence of values returned by the deletion operations:

i. Each consonant and the symbol ! corresponds to an addition of that letter onto the FIFO structure
ii. The symbol # and vowels correspond to a deletion operation on the FIFO structure, which removes and returns the earliest element of the structure.

b) Evaluate the given postfix expression using a stack. Provide a step-by-step explanation of each operation, including pushing operands onto the stack and applying operators until the final result is obtained. **[03]**

3, 5, +, 6, 4, -, *, 9, 3, /, +

c) Summarize the characteristics of a priority queue. **[02]**

2. **a)** Define stable and not stable sorting. Compare between linear search and binary search. **[03]**

b) A logistics company needs to arrange packages by weight in ascending order. Each package's weight is stored in a database, and swapping two weights in the database involves updating multiple records, which is time-consuming. However, comparing weights is a quick operation. The manager decides to use a sorting method that minimizes the number of swaps, even if it requires more comparisons. **[07]**

i. Based on the scenario, state which sorting algorithm is most suitable for this task. selection
ii. Implement the algorithm step by step.
iii. Assume the initial weights of the packages are [45, 12, 23, 89, 34, 88, 55]. Show the steps involved in sorting these weights using the chosen algorithm.

3. a) Here's a graph that is focused on identifying the traversing based on a graphical representation: **[05]**

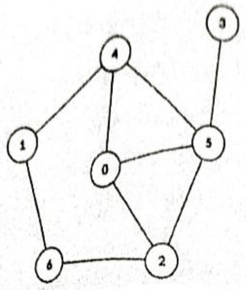

Figure 1: Undirected graph

i. Perform a Depth-First Search (DFS) starting from node 4. List the nodes in the order they are visited.
ii. Create the adjacency list for the given graph.

b) The row and column correspond to the nodes (A=0, B=1, C=2, D=3, E=4, F=5, G=6). **[05]**

$$\begin{bmatrix} 0 & 0 & 7 & 4 & 8 & 2 & 3 \\ 0 & 0 & 1 & 0 & 0 & 3 & 1 \\ 7 & 1 & 0 & 0 & 0 & 1 & 0 \\ 4 & 0 & 0 & 0 & 2 & 2 & 0 \\ 8 & 0 & 0 & 2 & 0 & 0 & 0 \\ 2 & 3 & 1 & 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Figure 2: Undirected weighted graph

i. Sketch the graph based on the above matrix.
ii. Apply the Kruskal's algorithm to find the minimum spanning tree of the given graph and show the total weight.

4. a) A tree has the property that each node has either 0 or 2 children. Determine what type of tree this is. Provide an example. Construct a Binary Search Tree (BST) by inserting the following sequence of numbers in the given order: [50, 30, 20, 40, 70, 60, 80, 23]. **[03]**

b) You are given two integer arrays preorder and inorder. Generate the binary tree.: **[02]**
Preorder Traversal: [3, 4, 6, 11, 9, 5, 8, 12, 13, 16, 20, 10]
Inorder Traversal: [11, 6, 4, 9, 3, 8, 13, 20, 16, 12, 5, 10]

c) You are given the word "FLOCCINAUCINIHHLILIFICION." The word contains multiple repeated characters with varying frequencies. **[05]**

i. Calculate the frequency of each character in the given text.
ii. Apply the Huffman coding algorithm to encode the word.
iii. Generate Huffman Codes.

5. a) Write the methods of hashing technique and explain it. [03]

b) A library is organizing a set of rare books into 10 secure lockers. Each book has a unique identification number, and the locker assignment is determined by a secret algorithm. However, if two books are assigned to the same locker, the library staff resolves it using a secondary algorithm to locate the next available locker. [05]

The books arrive in the following order with their identification numbers: 1043, 2022, 3008, 4129, 5096, 6041, 7083, 8037.

Secret Algorithm: h_o = key-value mod m;
Secondary Algorithm: $h_i = ((h_o+r)*k) \bmod m$;
where r = key-value mod p;

Here, m and p are prime numbers and p<m.

i. Identify the above mentioned technique. Explain the possible steps and reasoning behind following the above mentioned technique to achieve the final arrangement.
ii. Show the final arrangement after applying the secret algorithm to the book numbers.

c) Compare and illustrate the efficiency of Merge Sort algorithm against Insertion Sort algorithm in terms of time complexity. [02]